Lecture 7: Expectation, CDFs, PDFs and Problem Solving

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1 The Cumulative Distribution Function (CDF)

1.1 Intuition: Water Tank Analogy

The Cumulative Distribution Function (CDF) can be intuitively understood using a water tank analogy.

- Let the height h of the water in a cylindrical tank represent the value of a random variable.
- The volume of the water in the tank up to height h, denoted by $V(h)$, corresponds to the cumulative probability up to that value.

For a cylinder with radius R, the volume is given by:

$$V(h) = \int_0^h (\pi R^2) dh = (\pi R^2) h$$

Here:

- πR^2 is analogous to the Probability Density Function (PDF) of a uniform distribution.
- The maximum volume of the tank, $V(H) = \pi R^2 H$, is analogous to 1 (total probability).

1.2 Definition and Properties

Definition: The CDF of a random variable X is defined as:

$$F_X(x) = \Pr(X \leq x), \quad -\infty < x < \infty$$

Most of the information about the random experiment described by the random variable (RV) X is determined by the behavior of $F_X(x)$.

Properties of CDF:

1. **Bounds:** $0 \leq F_X(x) \leq 1$.
2. **Limits:**
$$F_X(-\infty) = 0, \quad F_X(\infty) = 1$$
3. **Monotonicity:** For $x_1 < x_2$,
$$F_X(x_1) \leq F_X(x_2)$$
(The CDF is a non-decreasing function).
4. **Interval Probability:** For $x_1 < x_2$, the probability that X lies in the interval $(x_1, x_2]$ is:
$$\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

1.3 Example #1: Valid CDFs

Determine which of the following functions are valid CDFs based on the properties (specifically boundary conditions and monotonicity):

1. $F_X(x) = \frac{1}{2} + \frac{1}{\pi}\tan^{-1}(x)$
   - **Valid.** Range is $[0, 1]$ as $x \to \pm\infty$.
2. $F_X(x) = [1 - e^{-x}]u(x)$
   - **Valid.** Standard exponential distribution form.
3. $F_X(x) = x^2 u(x)$
   - **Invalid.** As $x \to \infty$, $F_X(x) \to \infty$, violating the property $F_X(\infty) = 1$.
4. $F_X(x) = e^{-x^2}$
   - **Invalid.** As $x \to \infty$, $F_X(x) \to 0$, violating the property $F_X(\infty) = 1$.

1.4 Example #2: Calculations with CDF

Suppose a random variable has a CDF given by:

$$F_X(x) = (1 - e^{-x})u(x)$$

Compute the following probabilities:

1. **Probability** $\Pr(X > 5)$:

$$\Pr(X > 5) = 1 - \Pr(X \leq 5) = 1 - F_X(5) = 1 - (1 - e^{-5}) = e^{-5}$$

2. **Conditional Probability** $\Pr(X > 5 \mid X < 7)$:

$$\Pr(X > 5 \mid X < 7) = \frac{\Pr(5 < X < 7)}{\Pr(X < 7)} = \frac{F_X(7) - F_X(5)}{F_X(7)}$$

Substituting the CDF values:

$$= \frac{(1 - e^{-7}) - (1 - e^{-5})}{1 - e^{-7}} = \frac{e^{-5} - e^{-7}}{1 - e^{-7}}$$

2 The Probability Density Function (PDF)

2.1 PDF-CDF Relationship for Continuous Random Variables

The PDF of a random variable X evaluated at point x is defined as the limit:

$$f_X(x) = \lim_{\epsilon \to 0} \frac{\Pr(x \leq X < x + \epsilon)}{\epsilon}$$

Recall that for a continuous range:

$$\Pr(x \leq X < x + \epsilon) = F_X(x + \epsilon) - F_X(x)$$

Therefore, the PDF is the derivative of the CDF:

$$f_X(x) = \lim_{\epsilon \to 0} \frac{F_X(x + \epsilon) - F_X(x)}{\epsilon} = \frac{dF_X(x)}{dx}$$

Relationship Summary:

- The PDF is the derivative of the CDF:

$$f_X(x) = \frac{dF_X(x)}{dx}$$

- Conversely, the CDF is the integral of the PDF:

$$F_X(x) = \int_{-\infty}^{x} f_X(y)dy$$

2.2 Properties of PDF

1. **Non-negativity:** $f_X(x) \geq 0$.
2. **Normalization (Total Probability):** The total area under the PDF curve is 1.

$$\int_{-\infty}^{\infty} f_X(x)dx = 1$$

3. **Interval Probability:** The probability that X falls between a and b is the area under the curve from a to b.

$$\Pr(a < X \leq b) = \int_{a}^{b} f_X(x)dx$$

Graphically, $F_X(x_0)$ represents the area under the curve $f_X(x)$ from $-\infty$ to x_0.

2.3 Example #1: Valid PDFs

Identify valid PDFs from the following expressions:

1. $f_X(x) = e^{-x}u(x)$
   - **Valid.** $\int_0^\infty e^{-x}dx = 1$.
2. $f_X(x) = e^{-|x|}$
   - **Note:** $\int_{-\infty}^{\infty} e^{-|x|}dx = 2$. This requires normalization (multiplication by 1/2) to be a valid PDF.
3. $f_X(x) = \begin{cases} \frac{3}{4}(x^2 - 1), & |x| < 2 \\ 0, & \text{otherwise} \end{cases}$
4. $f_X(x) = \begin{cases} 1, & 0 \leq x < 1 \\ 0, & \text{otherwise} \end{cases}$
5. $f_X(x) = 2xe^{-x^2}u(x)$

2.4 Example #2: Derivations

From CDF to PDF: A random variable X has a CDF given by $F_X(x) = (1 - e^{-\lambda x})u(x)$. Its PDF is:

$$f_X(x) = \frac{dF_X(x)}{dx} = \frac{d}{dx}(1 - e^{-\lambda x}) = \lambda e^{-\lambda x}u(x)$$

From PDF to CDF: A random variable has a PDF given by $f_X(x) = 2xe^{-x^2}u(x)$. Its CDF is:

$$F_X(x) = \int_{-\infty}^{x} f_X(y)dy = \int_{0}^{x} 2ye^{-y^2}dy$$

Using integration techniques (substitution), the result is:

$$F_X(x) = (1 - e^{-x^2})u(x)$$

3 Expectation and Moments

3.1 Definitions

The expectation (or mean) is defined based on the type of random variable:

- **Discrete Random Variable (DRV):**

$$E[X] = \sum_k x_k P_X(x_k)$$

- **Continuous Random Variable (CRV):**

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

3.2 n^{th} Moments

The nth moment of a random variable X is defined as $E[X^n]$.

- **DRV:** $E[X^n] = \sum_k x_k^n P_X(x_k)$
- **CRV:** $E[X^n] = \int_{-\infty}^{\infty} x^n f_X(x) dx$

Specific Moments:

- $n = 0$**:** $E[X^0] = E[1] = 1$.
- $n = 1$**:** $E[X] = \mu_X$ (The Mean).
- $n = 2$**:** $E[X^2]$ (The Second Moment, used in variance calculation).

3.3 Central Moments

The nth central moment is defined as $E[(X - \mu_X)^n]$. This measures the deviation from the mean.

3.3.1 Variance ($n = 2$ Central Moment)

The variance σ_X^2 is the second central moment:

$$\sigma_X^2 = E[(X - \mu_X)^2]$$

Derivation of Variance Formula: Using the linearity of expectation:

$$\begin{aligned} E[(X - \mu_X)^2] &= E[X^2 - 2X\mu_X + \mu_X^2] \\ &= E[X^2] - 2\mu_X E[X] + \mu_X^2 \\ &= E[X^2] - 2\mu_X(\mu_X) + \mu_X^2 \\ &= E[X^2] - 2\mu_X^2 + \mu_X^2 \\ \sigma_X^2 &= E[X^2] - \mu_X^2 \end{aligned}$$

3.3.2 Higher Order Statistics

- **Skewness ($n = 3$):** Related to the 3rd central moment. Measures asymmetry.
- **Kurtosis ($n = 4$):** Related to the 4th central moment. Measures "tailedness".

3.4 Properties of Expectation

Expectation is a linear operation. For a constant a and random variable X:

- $E[X - \mu_X] = E[X] - E[\mu_X] = \mu_X - \mu_X = 0$ (The first central moment is zero).